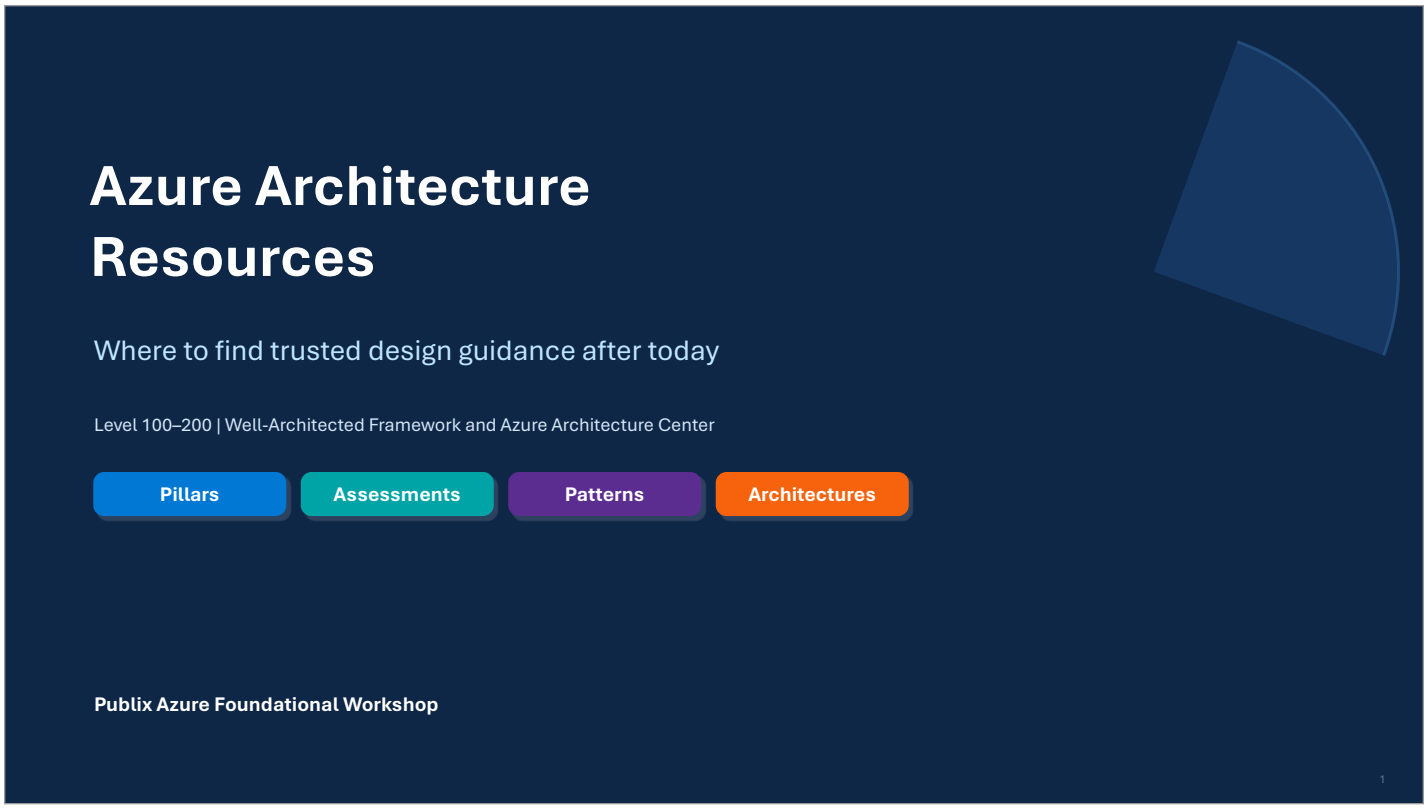
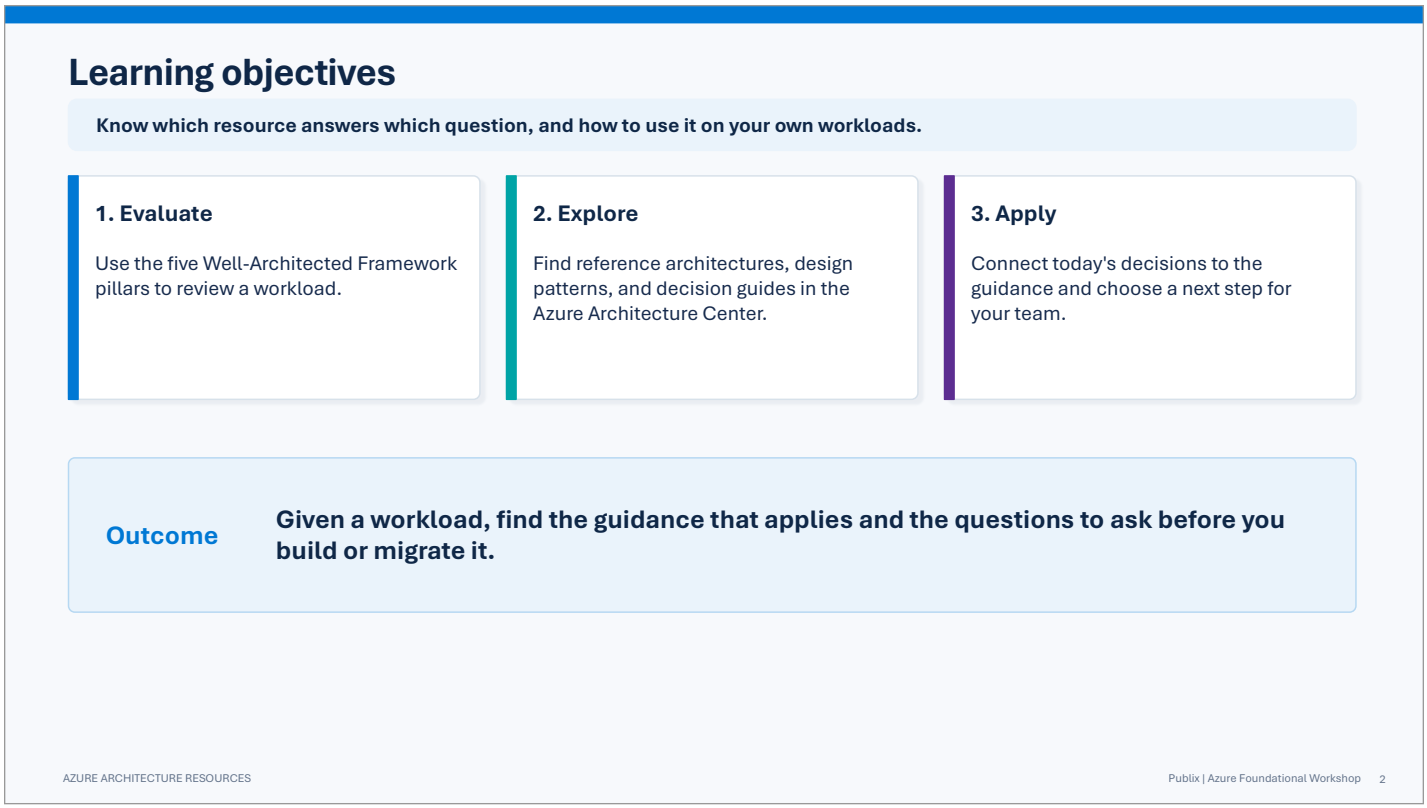



Slide 1 of 13

Welcome back. This closing session covers where to go after today. We'll look at two free resources on Microsoft Learn: the Azure Well-Architected Framework, which helps you judge whether a workload is designed well, and the Azure Architecture Center, which shows proven designs, patterns, and decision guides. There's no lab in this session. The goal is that everyone leaves knowing which resource to open for which question.

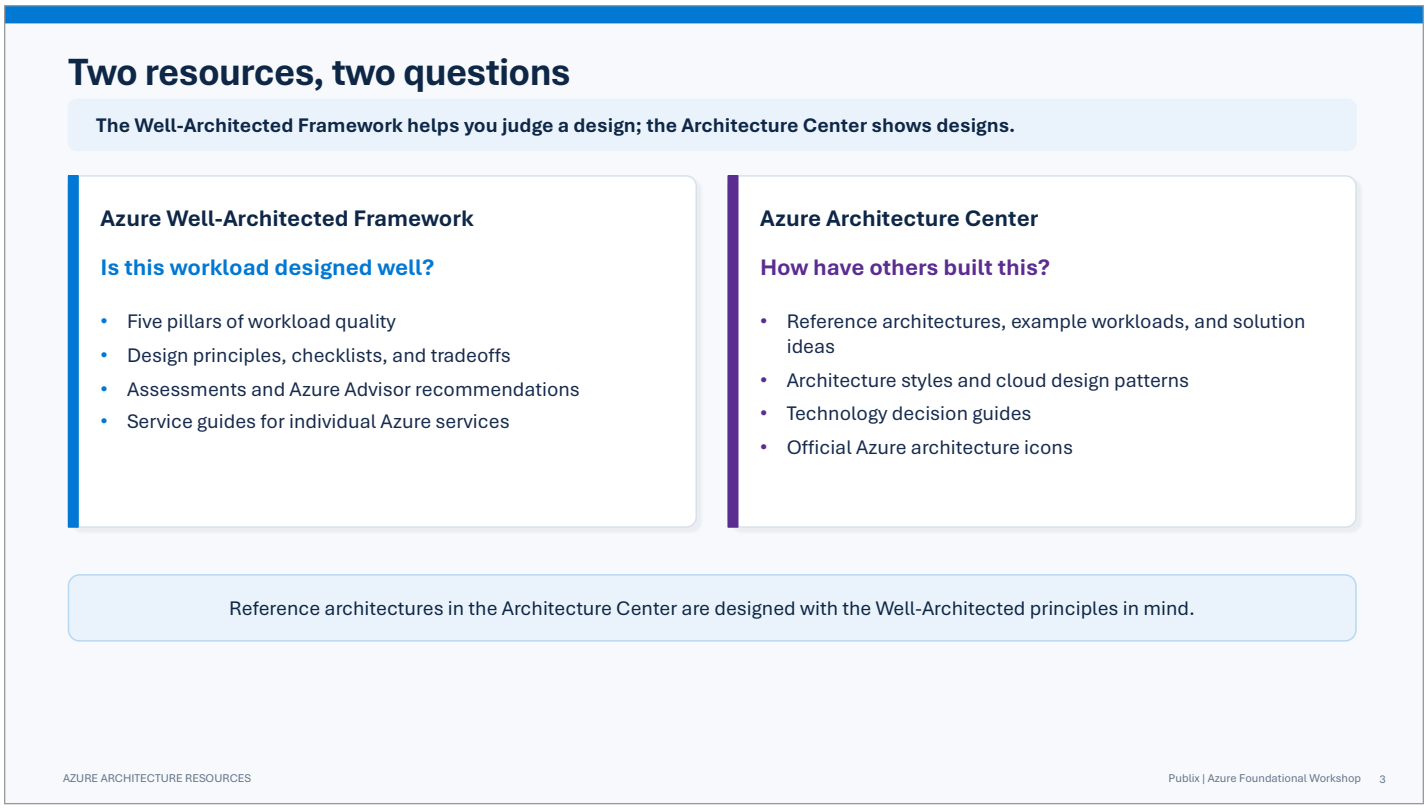
Learn more:
- Azure Well-Architected Framework: https://learn.microsoft.com/en-us/azure/well-architected/
- Azure Architecture Center: https://learn.microsoft.com/en-us/azure/architecture/




Slide 2 of 13

Three objectives. Evaluate: use the five pillars as a lens for any workload. Explore: know the main sections of the Architecture Center so you can quickly find a reference design or a decision guide. Apply: connect what we did today (placing, organizing, connecting, securing, building, and operating) to this guidance, and pick one concrete next step. Ask the room which workload they expect to migrate first, and use it as the running example.

Learn more:
- How to use the Well-Architected Framework: https://learn.microsoft.com/en-us/azure/well-architected/what-is-well-architected-framework
- Azure Architecture Center: https://learn.microsoft.com/en-us/azure/architecture/




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Position the two resources by the question each one answers. The Well-Architected Framework is a design framework: pillars, design principles, checklists, tradeoffs, and service guides that help you judge a workload. The Architecture Center is a catalog of solution ideas, example workloads, reference architectures, technology decision guides, and architecture guides. They work together: the reference architectures are designed with the Well-Architected principles in mind. If someone asks about organization-wide adoption and landing zones, that's the Cloud Adoption Framework, which the Publix platform team may already use.

Learn more:
- Well-Architected Framework pillars: https://learn.microsoft.com/en-us/azure/well-architected/pillars
- Azure Architecture Center: https://learn.microsoft.com/en-us/azure/architecture/
- Cloud Adoption Framework: https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/




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The framework is founded on five pillars: Reliability, Security, Cost Optimization, Operational Excellence, and Performance Efficiency. Each pillar provides design principles, a design review checklist, and documented tradeoffs. Stress balance: you can't maximize every pillar at once, so decisions are balanced against the business requirements. A useful habit is to ask the five questions on this slide in every design review.

Learn more:
- Pillars overview: https://learn.microsoft.com/en-us/azure/well-architected/pillars
- Reliability: https://learn.microsoft.com/en-us/azure/well-architected/reliability/
- Security: https://learn.microsoft.com/en-us/azure/well-architected/security/
- Cost Optimization: https://learn.microsoft.com/en-us/azure/well-architected/cost-optimization/
- Operational Excellence: https://learn.microsoft.com/en-us/azure/well-architected/operational-excellence/
- Performance Efficiency: https://learn.microsoft.com/en-us/azure/well-architected/performance-efficiency/

## The pillars in today's decisions

WELL-ARCHITECTED

**Today's decisions already map to the pillars; the framework gives them a shared vocabulary.**

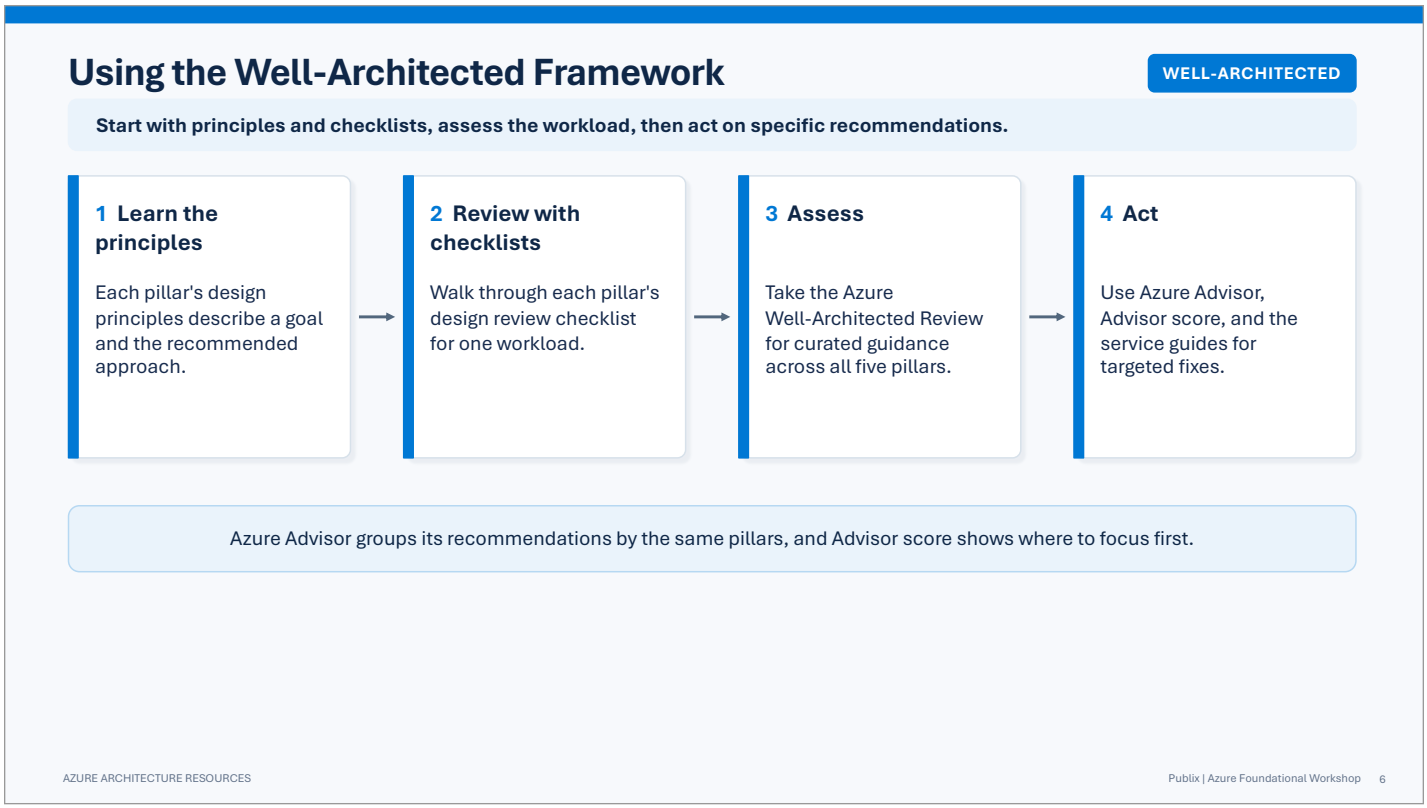
| Pillar | Where it showed up today |
|---|---|
| **Reliability** | Availability zones and regions; queues, retries, and dead-lettering |
| **Security** | Managed identities, Azure RBAC, Key Vault, and private endpoints |
| **Cost Optimization** | Tags, budgets, and Cost Analysis; choosing App Service plans and Flex Consumption |
| **Operational Excellence** | Infrastructure as code with Bicep; Azure Monitor alerts and runbooks |
| **Performance Efficiency** | Scale up and scale out; event-driven scaling with Azure Functions |



Slide 5 of 13

Walk the table and ask attendees to recall each item. Reliability: zones and regions this morning, plus the queue, retries, and dead-letter queue in Labs 2 and 3. Security: managed identities instead of connection strings, Azure RBAC at the narrowest scope, Key Vault, and private endpoints. Cost Optimization: tags, budgets, Cost Analysis, and choosing the right hosting plan. Operational Excellence: templates such as Bicep, plus alerts and runbooks. Performance Efficiency: scale up, scale out, and event-driven scaling. The point: they already think this way, and the framework gives them a shared vocabulary. The service guides apply the pillars to individual services.

Learn more:
- Service guides: https://learn.microsoft.com/en-us/azure/well-architected/service-guides/
- Architecture best practices for Azure App Service: https://learn.microsoft.com/en-us/azure/well-architected/service-guides/app-service-web-apps
- Architecture best practices for Azure Functions: https://learn.microsoft.com/en-us/azure/well-architected/service-guides/azure-functions
- Architecture best practices for Azure Service Bus: https://learn.microsoft.com/en-us/azure/well-architected/service-guides/azure-service-bus




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A practical sequence. First, read the design principles for each pillar. Second, use each pillar's design review checklist on a specific workload; each item is a key design strategy. Third, take the Azure Well-Architected Review, a self-assessment on Microsoft Learn that examines all five pillars and returns curated guidance. Fourth, act: Azure Advisor analyzes resource configuration and usage telemetry and groups its recommendations into categories that follow the pillars, and Advisor score helps you prioritize. The service guides add recommendations for individual services.

Learn more:
- Reliability design review checklist (example): https://learn.microsoft.com/en-us/azure/well-architected/reliability/checklist
- Azure Well-Architected Review: https://learn.microsoft.com/en-us/assessments/azure-architecture-review/
- Azure Advisor: https://learn.microsoft.com/en-us/azure/advisor/advisor-overview
- Advisor score: https://learn.microsoft.com/en-us/azure/advisor/advisor-score

## Every design choice is a tradeoff

WELL-ARCHITECTED

**Improving one pillar can cost another; name the tradeoff and decide deliberately.**

| Decision | Improves | Costs |
|---|---|---|
| **Add a second region** | **Reliability:** the workload survives a regional disruption | **Cost and operations:** more to pay for, deploy, and test |
| **Use private endpoints for every service** | **Security:** traffic stays on private addresses | **Operations:** more DNS and network design to manage |
| **Keep extra instances running** | **Performance and reliability:** headroom for demand peaks | **Cost:** you pay for capacity that sits idle |

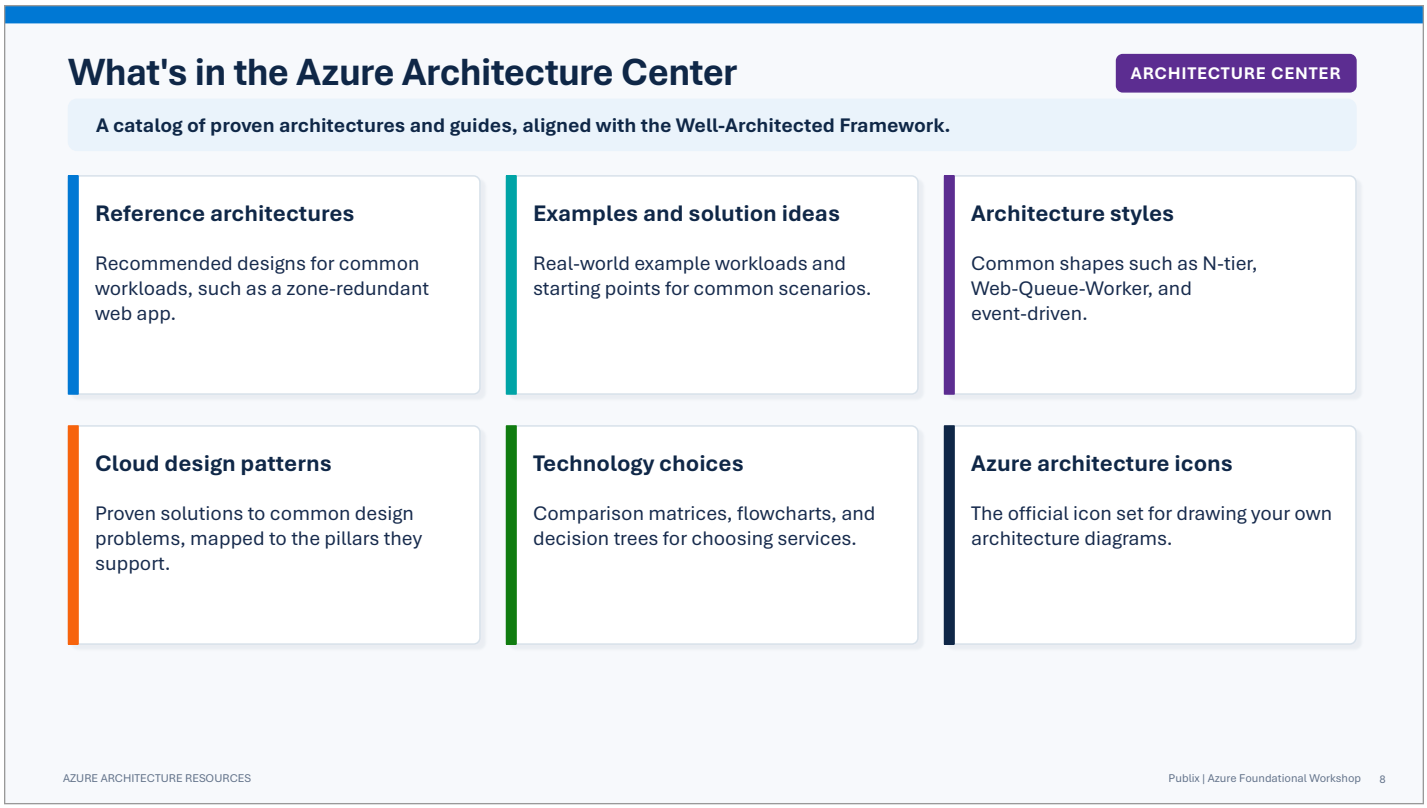
Each pillar documents its common tradeoffs, so teams can make the compromise on purpose.





Every architectural decision is a tradeoff, and the framework documents the common ones for each pillar. Use the three examples: a second region improves reliability but adds cost and operational work; private endpoints keep traffic on private addresses but add the DNS and network design you saw this morning; extra capacity helps performance and reliability, but you pay for idle instances. Ask: which of these would Publix accept for an order system, and who decides? Record the decision and the reason with the workload's design.

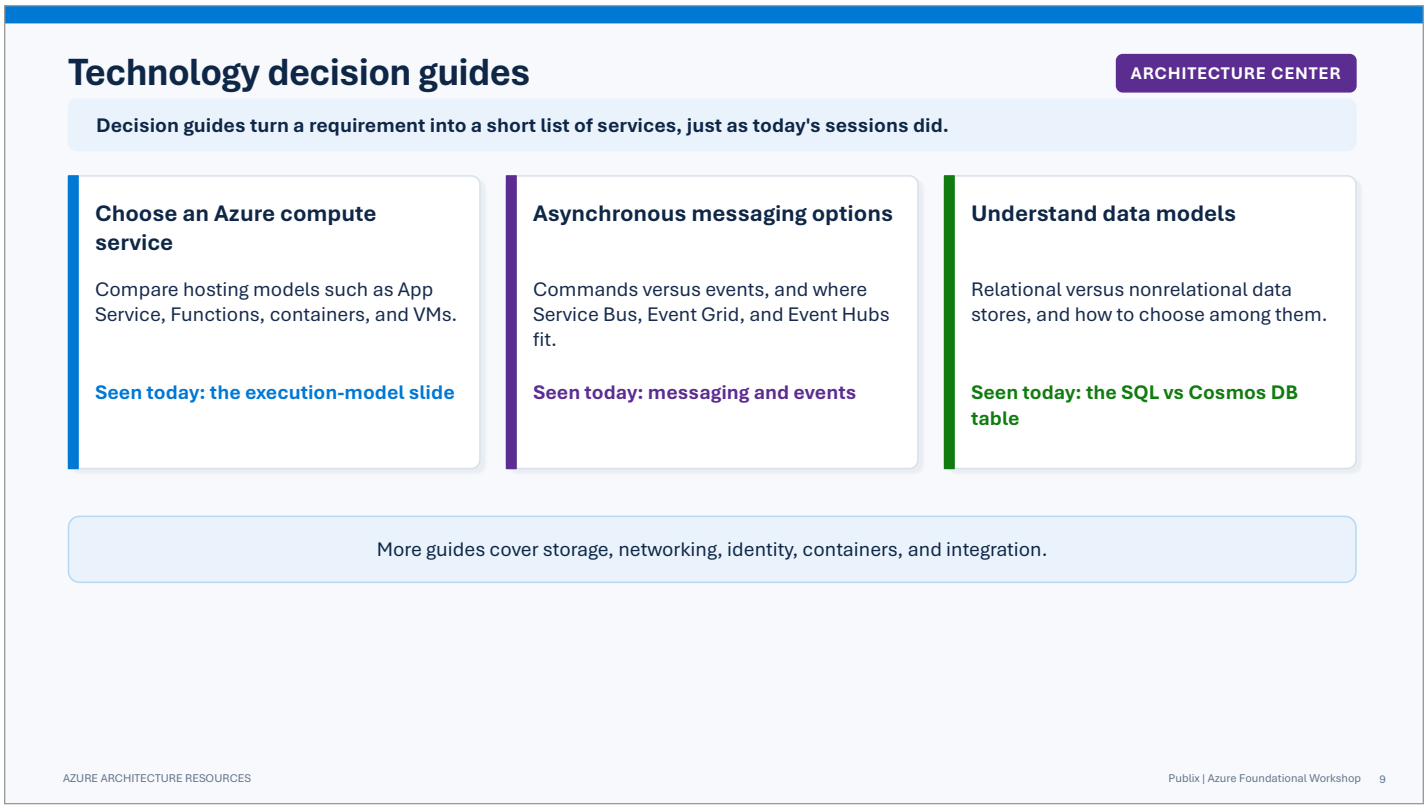
Learn more:
- Reliability tradeoffs (each pillar has a tradeoffs page): https://learn.microsoft.com/en-us/azure/well-architected/reliability/tradeoffs




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The Architecture Center is a catalog of solution ideas, example workloads, reference architectures, technology decision guides, and architecture guides. Show the entry points on its home page: browse architectures, learn the fundamentals, make technology choices, and apply design patterns. The browse page can be filtered by product, such as App Service or Service Bus. Mention the official Azure architecture icons for diagrams, and the guidance to put the product name near each icon.

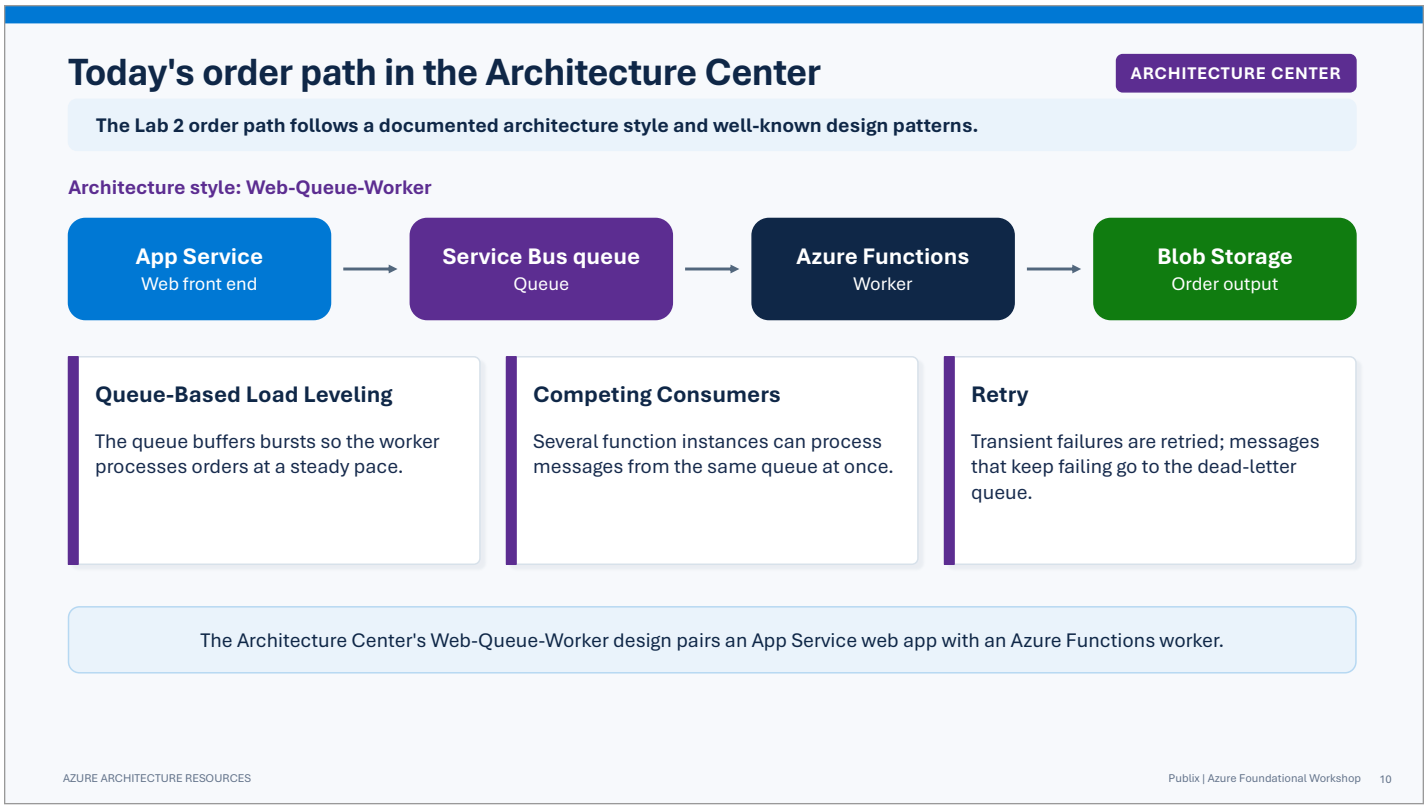
Learn more:
- Azure Architecture Center: https://learn.microsoft.com/en-us/azure/architecture/
- Browse architectures: https://learn.microsoft.com/en-us/azure/architecture/browse/
- Architecture styles: https://learn.microsoft.com/en-us/azure/architecture/guide/architecture-styles/
- Cloud design patterns: https://learn.microsoft.com/en-us/azure/architecture/patterns/
- Azure architecture icons: https://learn.microsoft.com/en-us/azure/architecture/icons/




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These guides use comparison matrices, flowcharts, and decision trees to narrow the options. The compute guide matches today's execution-model slide. The messaging guide explains commands versus events and where Service Bus, Event Grid, and Event Hubs fit. The data models guide compares relational and nonrelational stores, like the SQL Database versus Cosmos DB table. Encourage attendees to start with the guide, then confirm the choice with the Well-Architected service guide for that service.

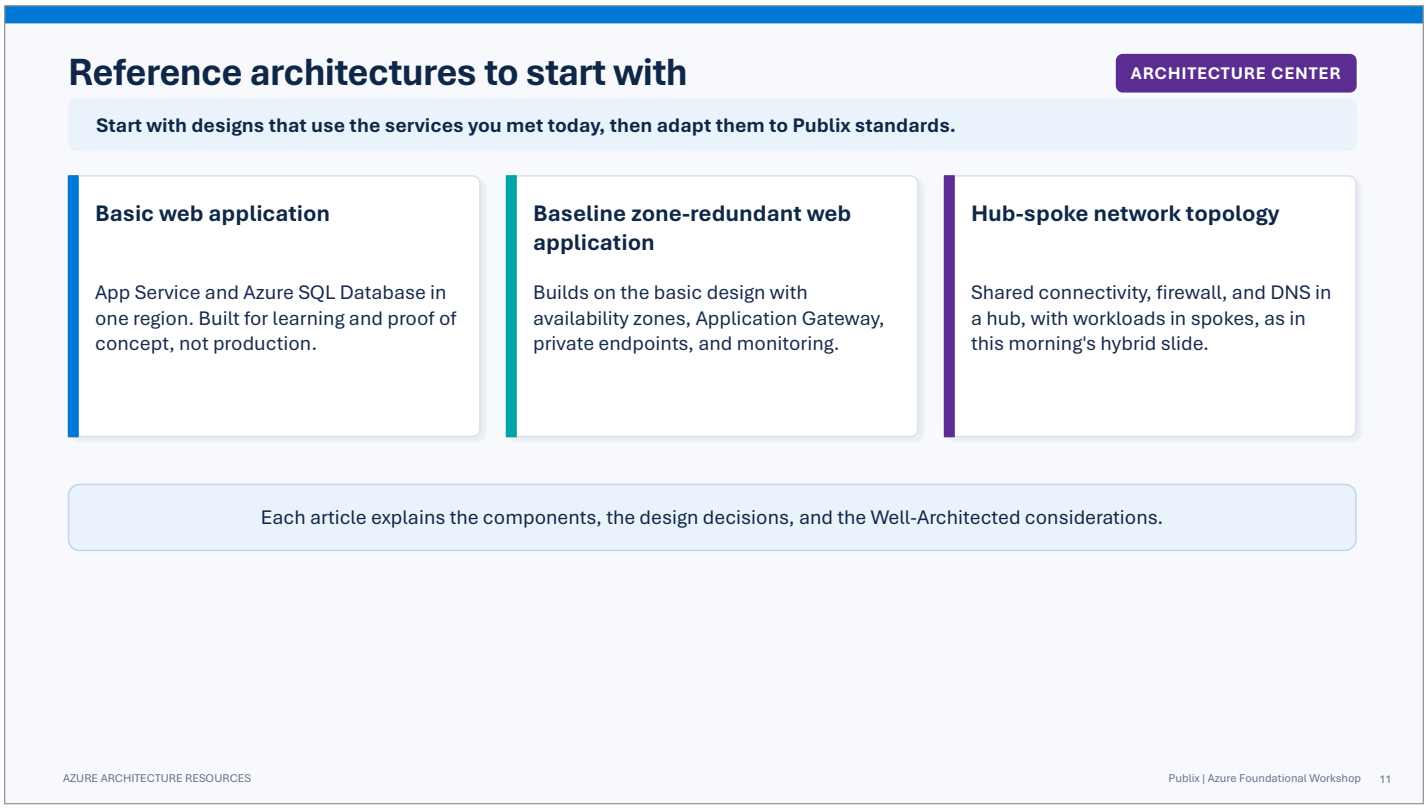
Learn more:
- Technology choices overview:
https://learn.microsoft.com/en-us/azure/architecture/guide/technology-choices/technology-choices-overview
- Choose an Azure compute service:
https://learn.microsoft.com/en-us/azure/architecture/guide/technology-choices/compute-decision-tree
- Asynchronous messaging options:
https://learn.microsoft.com/en-us/azure/architecture/guide/technology-choices/messaging
- Understand data models:
https://learn.microsoft.com/en-us/azure/architecture/data-guide/technology-choices/understand-data-store-models




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Bring back the Lab 2 order path. In the Architecture Center, this shape is the Web-Queue-Worker architecture style: a web front end handles client requests and hands longer work to a worker through a message queue. Its Web-Queue-Worker on App Service design uses an App Service web app as the front end and an Azure Functions app as the worker. Three patterns explain what attendees saw in the labs: Queue-Based Load Leveling, where the queue buffers bursts; Competing Consumers, where multiple instances process messages from the same queue; and Retry, for transient failures, with the dead-letter queue catching messages that keep failing.

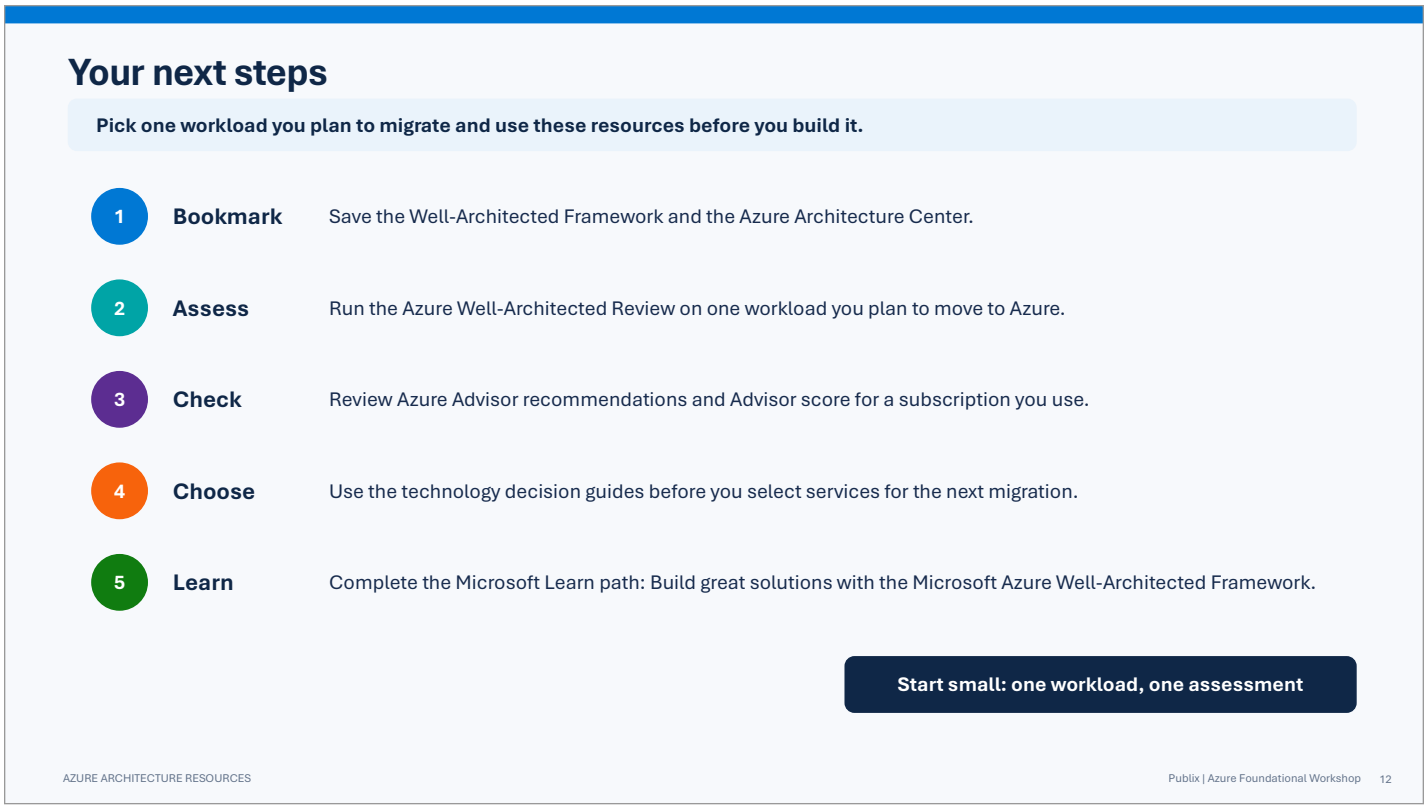
Learn more:
- Web-Queue-Worker architecture style: https://learn.microsoft.com/en-us/azure/architecture/guide/architecture-styles/web-queue-worker
- Queue-Based Load Leveling pattern: https://learn.microsoft.com/en-us/azure/architecture/patterns/queue-based-load-leveling
- Competing Consumers pattern: https://learn.microsoft.com/en-us/azure/architecture/patterns/competing-consumers
- Retry pattern: https://learn.microsoft.com/en-us/azure/architecture/patterns/retry




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Three good starting points. The basic web application is an introductory setup for learning and proof of concept, not production. The baseline highly available zone-redundant web application builds on it with a secure, zone-redundant design that adds Application Gateway, private endpoints, a virtual network, and monitoring. Hub-spoke is the network topology behind this morning's hybrid connectivity slide. Remind attendees to adapt any reference design to Publix standards and to check with the platform team.

Learn more:
- Basic web application:
https://learn.microsoft.com/en-us/azure/architecture/web-apps/app-service/architectures/basic-web-app
- Baseline highly available zone-redundant web application:
https://learn.microsoft.com/en-us/azure/architecture/web-apps/app-service/architectures/baseline-zone-redundant
- Hub-spoke network topology:
https://learn.microsoft.com/en-us/azure/architecture/networking/architecture/hub-spoke




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Close with one concrete commitment per person or team. The Well-Architected Review works best with a real workload in mind, ideally one Publix plans to migrate. Advisor needs a subscription with real resources. The Microsoft Learn path Build great solutions with the Microsoft Azure Well-Architected Framework is a beginner-level path of six modules. If the platform team is preparing landing zones, Microsoft Assessments also includes an Azure Landing Zone Review.

Learn more:
- Azure Well-Architected Review: https://learn.microsoft.com/en-us/assessments/azure-architecture-review/
- Azure Advisor: https://learn.microsoft.com/en-us/azure/advisor/advisor-overview
- Technology choices: https://learn.microsoft.com/en-us/azure/architecture/guide/technology-choices/technology-choices-overview
- Learning path: https://learn.microsoft.com/en-us/training/paths/azure-well-architected-framework/

## Resource links

**Bookmark these links; everything here is free on Microsoft Learn.**

| Resource | Use it to | Link |
|---|---|---|
| **Azure Well-Architected Framework** | Learn the pillars, principles, and checklists | learn.microsoft.com/azure/well-architected |
| **Azure Well-Architected Review** | Assess a workload against the five pillars | learn.microsoft.com/assessments/azure-architecture-review |
| **Azure Advisor** | See recommendations for your Azure resources | learn.microsoft.com/azure/advisor/advisor-overview |
| **Azure Architecture Center** | Browse architectures, guides, and patterns | learn.microsoft.com/azure/architecture |
| **Cloud design patterns** | Solve common cloud design problems | learn.microsoft.com/azure/architecture/patterns |
| **Technology choices** | Compare and choose Azure services | learn.microsoft.com/azure/architecture/guide/technology-choices/technology-choices-overview |
| **Azure architecture icons** | Draw your own architecture diagrams | learn.microsoft.com/azure/architecture/icons |



Slide 13 of 13

Leave this slide up during questions, or share the deck afterward. Every resource here is on Microsoft Learn and free to use.

Learn more:
- Azure Well-Architected Framework: https://learn.microsoft.com/en-us/azure/well-architected/
- Azure Well-Architected Review: https://learn.microsoft.com/en-us/assessments/azure-architecture-review/
- Azure Advisor: https://learn.microsoft.com/en-us/azure/advisor/advisor-overview
- Azure Architecture Center: https://learn.microsoft.com/en-us/azure/architecture/
- Cloud design patterns: https://learn.microsoft.com/en-us/azure/architecture/patterns/
- Technology choices: https://learn.microsoft.com/en-us/azure/architecture/guide/technology-choices/technology-choices-overview
- Azure architecture icons: https://learn.microsoft.com/en-us/azure/architecture/icons/